

Touch-Sensitive Hand Interactions for Social Robots Using Fiber Bragg Grating Sensors

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Abstract—Physical human-robot interaction (pHRI) has been demonstrated to be essential in the implementation of social assistive robots (SARs), which require advanced sensing capabilities for accurate and responsive engagement. This study presents the development and validation of a fiber Bragg grating (FBG) sensor network integrated into the hand of the CASTOR robot to classify complex pHRI. Nine pHRI were collected and evaluated within the high five, pets, handshakes, hits, and pinches categories. Four machine learning (ML) algorithms were tested, and the Bagged Decision Tree Classifier (BDTC) achieved the best performance. During testing, the model achieved an accuracy of 98%. The results demonstrate that the proposed FBG sensor network can classify complex pHRI. Future work will explore additional instrumented areas of the robot and expand the physical interaction analysis to enhance social robot adaptability and user experience.

Index Terms—Physical Human-Robot Interaction; Sensor-based Control; Social Robots.

I. INTRODUCTION

Social Assistive Robotics (SARs) support healthcare scenarios through cognitive and physical interactions [1]. These robots have been implemented in cardiac rehabilitation and elderly care, demonstrating their potential to improve patient outcomes [2]. In addition, SARs have emerged as a transformative tool in the treatment of neurodevelopmental disorders, such as Autism Spectrum Disorder (ASD) [3]. In particular, ASD is characterized by challenges in social communication and interaction, as well as repetitive sensory-motor behaviors [4]. ASD is a lifelong condition, and early diagnosis is essential for improving quality of life [5]. Traditional interventions include joint attention, behavioral management, and cognitive behavioral therapies [6]. These approaches enhance emotional regulation, cognitive abilities, and daily life skills [3], [5]. However, maintaining engagement in therapy remains a challenge.

SAR systems have demonstrated the capability to reduce resistance to social interaction [7], increase eye contact [8],

improve body awareness through imitation [6], and stimulate communication skills [9]. Furthermore, these robots have been deployed to encourage physical interaction, a fundamental aspect of human well-being [10]. For instance, robots such as Probo [11], KASPAR [10], and Paro [12] have been implemented for this purpose. However, the use of these devices in clinical settings has been limited by high costs and restricted accessibility. In addition, more complex robots (i.e., NAO) generate concerns due to their fragility and the side damage caused by their actuators [13]. Thus, developing social robots equipped with compliant systems is essential to enable an effective touch-sense interaction.

In this context, the CASTOR robot has addressed these limitations through innovative design, integrating flexible soft robotics and elastic actuators [14]. The device was projected to provide natural physical human-robot interaction (pHRI). Its flexibility, huggable structure, and shock-absorbing mechanisms ensure the robot's durability and users' safety, making it suitable for therapeutic use with individuals who may exhibit unpredictable behaviors [14].

Besides, the use of tactile and pressure sensors has been explored to monitor pHRI in diverse scenarios. As presented by Gaitán-Padilla *et al.*, these sensors allowed CASTOR to provide more natural hugs compared with time-controlled interactions [15]. Likewise, sensors are used to feed ML models to classify between different states, as in the design of an autonomous huggable robot through the detection of intra-hug gestures [16]. This ability ensured a more personalized experience for users and also highlighted the robot's adaptability in social contexts. However, more sensitive sensors are needed to estimate more complex pHRI, mainly those of specific anatomic structures (e.g., the hand) [17].

Optical fiber-based sensors, particularly fiber Bragg grating (FBG) sensors, have emerged as a promising alternative due to their high sensitivity and precision, immunity to electromagnetic interference, fatigue resistance, and multiplexing capabilities [18]. In [19], the development of a flexible bionic skin based on FBGs is presented to detect pressure, slip, and ambient temperature. The innovation lies in the design of a double-layer distribute array, which increases tactile sensing accuracy and enables the identification of sliding speed. Similarly, [20] introduces a smart skin capable of detecting the intensity and location of applied forces.

The sensitivity of the FBG sensors in detecting touch and force variations is strengthened by their application in various

This work is partially supported by FAPES (EXTRATO ATA DA 6ª REUNIAO CCAF/2022 – “CPID 2.2”, 633/2022), CNPq (310668/2021-2, 404111/2023-8, and 402739/2024-8), Finep (Tec assistiva 2 - 2132/22), and IEEE RAS SPARX.

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intricate robotic domains to develop more intelligent systems, including soft robotic grippers [21] and HRI systems [22]. In this sense, D'Abbraccio *et al.* integrated an artificial skin into the forearm of a collaborative robot [23], mapping the interactions and enhancing intuitive behaviors with the users, laying the groundwork for the use of FBG sensors in SAR applications. In this sense, the main contribution of this work is the development and validation of an FBG-based sensor adapted to the CASTOR robot's hand to identify complex pHRIs. In this way, ML algorithms were tested to classify high-five, pet, handshaking, hit, and pinch interactions. This approach aims to generate real-time feedback within therapeutic sessions, fostering assertive cause-and-effect behaviors.

II. METHODOLOGY

This section presents the developed FBG sensor and the adaptation to the robot's hand. It describes the setup proposed to identify the physical interactions, the calibration stage, the experimental procedure, data acquisition, and signal processing. Additionally, it covers the training and validation of different ML algorithms such as k-nearest neighbor (KNN), support vector machine (SVM), bagged decision tree classifier (BDTC), and a convolutional neural network (CNN), used in other interaction identification works [24].

A. Robotic Platform

The CASTOR robot¹ is an affordable, open-source platform designed to facilitate pHRIs for children with ASD. The mechanical design incorporates two novel motion mechanisms, elastic actuators and bioinspired tendons, to ensure the user safety. The robot is equipped with seven motors (Dynamixel MX/AX, Robotis, France), five servomotors (MG996, Towerpro, Taiwan), and a passive damping system resulting in 14 degrees of freedom (DoF). Regarding the dynamic functionalities, the system include arm and neck movement, emotions expresion, and the internal huggable system. Besides, CASTOR is a robust 3D-printed modular device, easy to build and repair. The robot's appearance is shown in Figure 1.

B. FBG Sensing Principle

The operating principle of an FBG is based on the generation of a distributed Bragg reflector into an optical fiber, usually using ultraviolet (UV) phase light masks. This is a periodic modulation of the refractive index along a short segment of the fiber's core. This grating reflects a specific wavelength peak (λ_{Bragg}) when is connected to a broadband light source [25]. The λ_{Bragg} central wavelength depends on the grating period in the core (Λ) and the refractive index n_{eff} as defined in Equation 1.

$$\lambda_{Bragg} = 2n_{eff}\Lambda \quad (1)$$

FBGs are sensitive to temperature and mechanical deformation [25] which makes them useful for sensing applications by monitoring the λ_{Bragg} shifts, indicating changes in

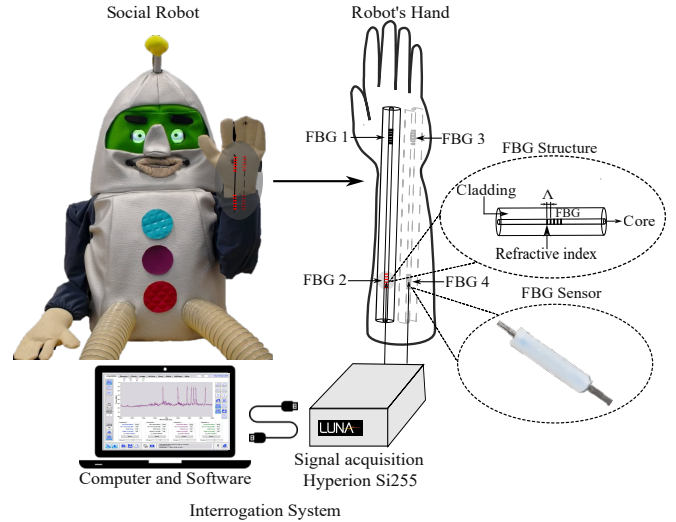


Fig. 1: Sensing system setup. The FBG network into the robot's hand and the interrogation system for data acquisition.

external conditions. The FBG structure is illustrated in Figure 1, which depicts the FBG and the interrogation system used.

For strain, the central wavelength of the FBG shifts in response to the induced perturbation. The relative Bragg wavelength shift ($\Delta\lambda_B/\lambda_{B0}$), caused by strain ($\Delta\epsilon$) and temperature variations (ΔT), is described by Equation 2. In this equation α and ξ represent the thermal expansion and thermo-optic coefficients, respectively, while ρ_ϵ denotes the effective photo-elastic coefficient.

$$\frac{\Delta\lambda_B}{\lambda_{B0}} = (1 - \rho_\epsilon)\Delta\epsilon + (\alpha + \xi)\Delta T \quad (2)$$

C. FBG Sensor Network

The proposed FBGs were fabricated using an UV laser and phase mask technique. This process involves inscribing the desired periodic refractive index modulation along the core of a single-mode optical fiber [26]. The process consisted of preparing two segments of boron-doped photosensitive optical fiber (FiberCore, United Kingdom), and removing the coating from the designated FBG inscription section to enable efficient UV exposure. Subsequently, two-phase masks (Ibsen Photonics, Denmark) were used to inscribe the λ_{Bragg} in the exposed core section.

The UV laser (CNI DPS-266, China), operating at 266 nm and emitting pulses of 8 ns at a repetition rate of 100 Hz was employed to create the interference pattern [27]. The alignment ensured uniform exposure, while the laser exposure time was controlled with a optical sensing interrogator SM125 (Micron Optics, Inc.). After inscription, the fibers were coated with DeSolute 950-200 (Fiber Optic Center, Inc., United States) for protection, using a Vytran Fiber Recoater (Thorlabs, United States). Finally, a refractive-index gel was applied to the end of the fiber to reduce undesirable light reflection.

The inscribed FBGs use 3D-printed guides made of Poly-lactic acid (PLA) to facilitate their integration into the robot

¹CASTOR robot repository.

hand. In addition, it has been demonstrated the need to enhance its mechanical resistance, embedding them into smart textiles [21], flexible 3D-printed structures [28], or viscoelastic materials [23]. For that reason, the sensitive areas of the proposed network were covered with a layer of silicone (EcoFlex, Polisil, Brazil) also increasing the sensing area and improving sensitivity. This encapsulation of the FBG sensor is depicted in Figure 1.

Each segment of the robot's hand was instrumented with a dedicated FBG sensor, resulting in four multiplexed FBGs. As shown in Figure 1 FBG1 and FBG2 were located at the front, while FBG3 and FBG4 were placed at the back. Moreover, the centered λ_{Bragg} for FBG1 and FBG3 was 1548.3 nm, while for the remaining FBGs, it was 1562.6 nm.

D. Sensors' Network Calibration

The four FBG sensors in the network were calibrated using a precision force gauge. A controlled force ranging from 1N to 5N and back to 1N was applied in 1N increments, with each force level maintained for 15s. This cycle was repeated five times to ensure repeatability and robustness. The collected data were then used to compute the sensors' sensitivity, evaluate linearity and hysteresis behavior, and analyze measurement deviation across the force range. The Linear Regression (LR) of the obtained calibration was used to identify the relationship between the wavelength shift and the exerted forces.

E. Experimental Procedure

Regarding data collection, two volunteers (females, 24 ± 1 years old) made 10 recordings of three interactions of each type in 5 seconds, resulting in a total dataset of 270 segments. Additionally, data were recorded at a sampling rate of 1000 Hz using an interrogator (Hyperion Si255, LUNA, Inc., United States) and the ENLIGHT Software (Micron Optics, Inc., United States). Figure 2 illustrates how each physical interaction was performed.

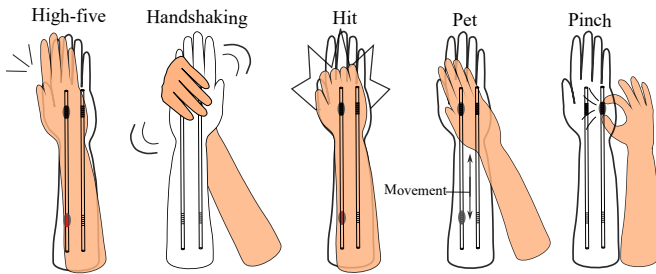


Fig. 2: Main pHRI conducted with the robot's hand during the experimental procedure.

It is important to highlight that the five main interactions were high-five, handshaking, hit, pet, and pinch. However, for pets, both frontal and back versions were performed. As for pinching, one interaction was proposed for each sensor, resulting in four distinct classes for this pHRI. In total, nine categories were obtained.

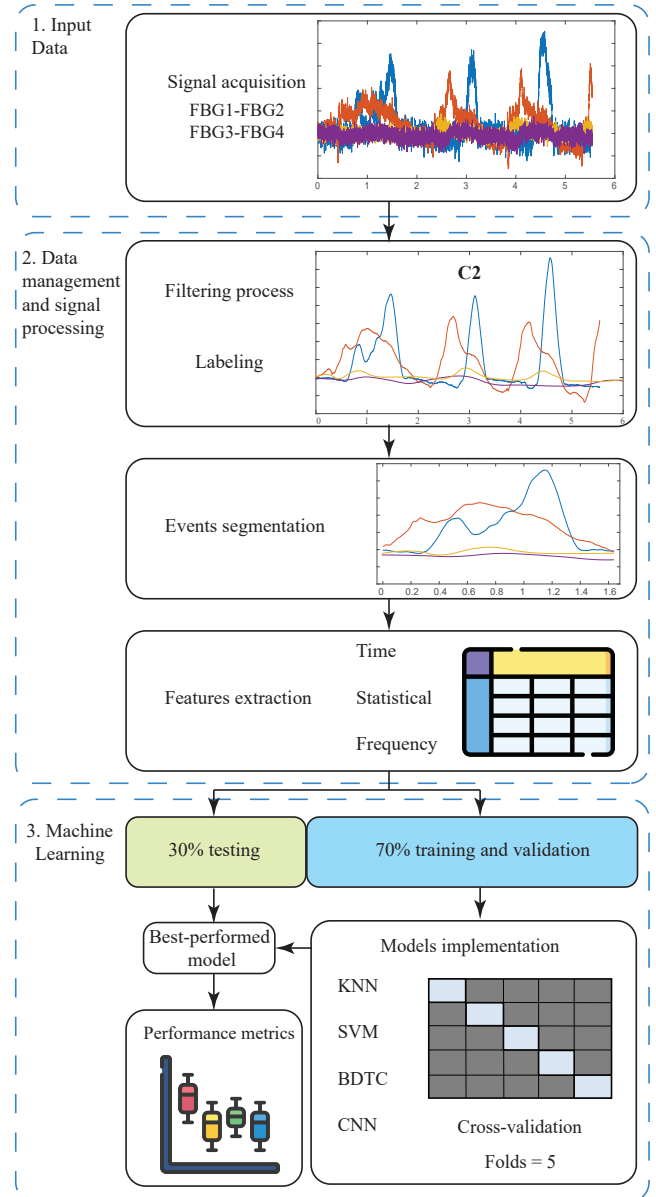


Fig. 3: Flux diagram for the features extraction and data flow for the pHRI classification.

F. Data Analysis

The dataset was divided into 70% for training (216 samples) and 30% for testing (54 samples), following a standard approach to ensure the model retains sufficient data for accurate evaluation. Feature extraction was performed on the signals, incorporating frequency-domain and spatiotemporal features such as interaction duration, maximum, minimum, mean, median, root mean square, standard deviation, skewness, kurtosis, and the number of detected peaks. Subsequently, several ML algorithms, including K-NN, SVM, BDTC, and CNN [24] were trained and compared. The model configurations were as follows: Euclidean distance with $k=5$ for K-NN, a linear kernel for SVM, six maximum splits and 30 learners for BDTC, and a two-layer CNN with the first layer comprising 25 units and a ReLU activation function. A 5-fold cross-validation methodology was used

to evaluate the performance of each algorithm and identify the best-performing model. The complete data management, processing, and ML pipeline is illustrated in Figure 3.

III. RESULTS AND DISCUSSION

The characterization phase of the sensors' responses to applied forces, the LR is presented in Figure 4, highlighting the linearity and sensitivity of the wavelength shifts for each sensor in the network. The optical interrogator has a resolution of 1 pm, thus, the sensors can measure a maximum resolution of 33 mN. In addition, the sensitivity obtained was 0.03 nm/N, demonstrating a high responsiveness to mechanical interactions, even when the sensors were covered with silicone and the soft foam padding of the robot's hand. The hysteresis observed for each sensor remained within the standard deviation across the five trials, indicating minimal lag between loading and unloading cycles. Finally, the linearity of the sensors is confirmed by the correlation between the measured points and the fitted LR (R^2), demonstrating a consistent calibration to applied forces.

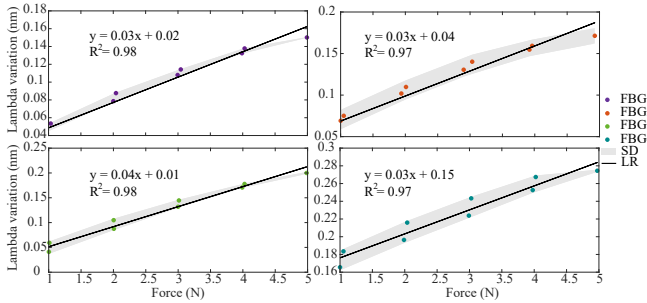


Fig. 4: Characterization and calibration of the FBG sensor network in response to applied force variations.

The time-domain response of each pHRI was analyzed to assess the behavior of the FBG sensors during different interaction types. Figure 5 displays the response patterns of the four FBGs during the five primary pHRI scenarios. For the petting and pinching interactions, representative cases were selected to illustrate the characteristic signal behavior. The resulting signals demonstrate that different pHRI types generate different activation profiles depending on the nature of the interaction. For instance, cases (A) and (C) showed more peaked behavior, while (B), (D), and (E) produced smoother, Gaussian-like distributions. Following the calibration phase, the wavelength shifts observed in (A) and (B) ranged from -0.02 to 0.10 nm, while those in (D) and (E) remained below 0.15 nm. Notably, case (C) presented higher variations, with shifts reaching approximately 0.60 nm, significantly greater than those observed in the other interaction types. These findings highlight the system's ability to detect and distinguish between different types of complex pHRI.

The results indicate a consistent relationship between the pHRI and FBG activation. This coherent behavior has been explored and demonstrated in the calibration phase through the characterization with applied forces and in previous studies. Li *et al.* proposed the use of FBG-based tactile sensors

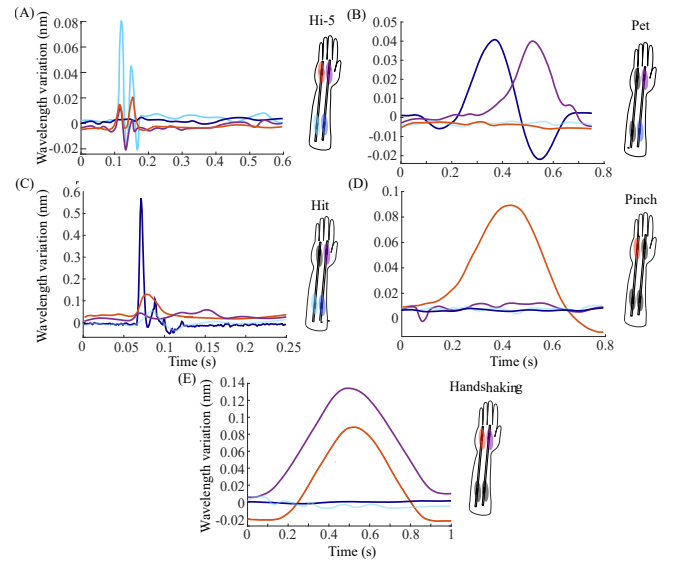


Fig. 5: Time-domain signals for each pHRI, showing the activation and deactivation of the affected FBGs.

for robotic fingers [22], enabling the precise identification of small forces and the recognition of complex textures. Besides, more complex and reduced-area estimations of robot-related variables using FBGs have been conducted, including localized complex interactions and concentrated contact trends [29].

Furthermore, the time-domain analysis suggests that the FBG network detects instantaneous variations. The system tracked interactions ranging from 1-second (handshaking) to 0.05-second (hits), capturing dynamic time characteristics. This capability is relevant for pHRI identification and SAR implementations in ASD therapies, where engagement-based contact is common [30].

On the other hand, to validate the sensing system and to identify the nine pHRI types, were implemented four classification algorithms commonly used for state estimation [31]. The performance metrics for each model are presented in Table I. The BDTC demonstrated the highest classification accuracy (i.e., 98%) across the nine interaction classes. This aligns with its previous applications in physiological estimation [31]. These results highlight the effectiveness of BDTC algorithms in signal-based classification tasks, particularly in recognizing complex touch gestures.

After selecting the BDTC model, its performance was evaluated using the remaining 30% of the dataset. During this stage, each pHRI was assessed individually to calculate the four performance metrics. This method enabled a detailed analysis of the variability in accuracy, precision, recall, and F1 score, presented in Figure 6.

The results demonstrate the model's robustness, achieving a median accuracy of 98%. However, precision and recall varied across specific pHRI categories, suggesting limitations in detecting localized or low-intensity gestures. This is evident in the low recall for pinches (66.7% in FBG 2) and the 75% precision for forehead pets. These gestures appear to generate overlapping signal patterns in

amplitude and duration, making them harder to distinguish. This challenge may relate to the BDTC model's structure, which does not inherently handle time-based data or capture signal progression, limiting its ability to differentiate gestures with similar temporal profiles [32]. In addition, the soft structure of the robot could generate a mechanical limitation to low amplitude variations (lower than 0.03 nm), generating similarities between interaction types.

In line with these findings, the observed discrepancies in recognition rates among the different pHRI categories in Figure 6 suggests that some gestures, such as high-five and handshake, which exhibit distinct pressure and duration patterns, were more accurately classified than pinch variants, which share overlapping sensor activations. Similarly, O'Connell *et al.* identify that the pinch interactions were the most difficult to classify across the three regions, with 95% on the head, 65% on the cheeks, and 85% on the head [33].

These results highlight the importance of refining the sensing strategies to capture variations during physical interactions. Although the current FBG network configuration provides high sensitivity and robustness, the system's ability to differentiate these gestures could be enhanced by increasing the spatial resolution of the sensor distribution, incorporating advanced temporal feature fusion techniques, and modifying the mechanical interface to amplify microdeformations.

TABLE I: Comparison of classification algorithms performance metrics for pHRI recognition.

Algorithm	Accuracy	Precision	Recall	F1-score
K-NN	0.97	0.85	0.84	0.85
SVM	0.97	0.89	0.88	0.89
BDTC	0.98	0.89	0.89	0.89
CNN	0.96	0.84	0.84	0.84

Comparing the obtained results, this work showed a more intricate classification by using nine interactions and obtaining better accuracies in the complex ones. Knight *et al.* identified four pHRI and obtained 100% of accurate region-touched detection in foot and head. Moreover, the complex pHRI of hugging and thistle obtained lower accuracies (80% and 20%, respectively) [34]. Similarly, O'Connell *et al.* achieved 100% accuracy in detecting contact and pet interactions on the head and trunk regions, while the cheek region had lower accuracies (70%, 60%, 50%, 55%, and 65%) [33]. These findings reinforce the complexity of natural pHRI and highlight the need for robust sensing, processing, and classification techniques.

Integrating this system into SAR platforms, like CASTOR, enhances intelligence and natural interaction capabilities. This work contributes to incorporating advanced sensors in affordable robots to enable feedback-inspired interactions [35]. Particularly, in the case of the CASTOR robot, this application could improve the perception and therapy flux with children with ASD, giving optional capabilities to the robot, such as the addition of cause-effect responses and interactive behaviors. Additionally, the correct implementation, along with the high sensitivity and resistance to different

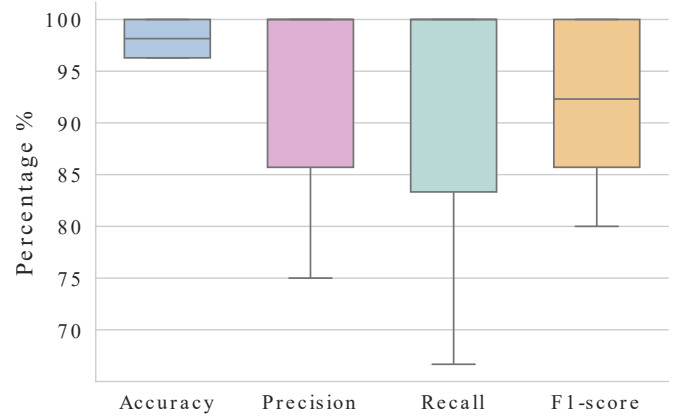


Fig. 6: Boxplot of the performance metrics evaluating the best-performed BDTC algorithm with the testing data.

interactions, highlights the adaptability of the proposed sensor. The robustness of the robot and the sensing systems is essential to ensure effective and adaptable interventions [36].

IV. CONCLUSION AND FUTURE WORK

This study presented the development and validation of an FBG-based sensor to identify complex pHRI. A four-FBG network system was fabricated and integrated into the robot's hand, embedded in a 3D-printed structure with a silicone layer. A total of 270 interactions were captured, including high-fives, pets, handshakes, hits, and pinches. The results demonstrated that the FBG network identified nine interactions using ML algorithms. In particular, the BDTC model achieved the highest accuracy, reaching 98%. This approach highlights the potential of the proposed system to enhance robot sensitivity, enabling more natural and socially meaningful interactions, particularly in therapeutic applications for individuals with ASD. However, some limitations were observed in precision and recall for pinches and pets in the test-phase of the algorithm, indicating the need for improved sensor sensitivity or algorithm adjustments.

In a specific and essential pHRI, Block *et al.* highlight the necessity of precise gesture recognition in affective hugging interactions [16]. They demonstrated how sensors contribute to the classification of four intrahug gestures (hold, rub, pat, and squeeze), highlighting the importance of accurate classification to avoid negative perceptions toward the robot's intelligence. The results of this study show a high performance in the detection of the nine classes, following the main concern about the correct feedback behavior of the robot. In addition, this is aligned with the previously validated capability of the CASTOR robot to deliver sensor-controlled reciprocal hugs, leveraging its 14 DoF, soft external structure, and huggable internal damping system [15]; which could be further improved through the expansion of the network.

Similarly, it has been examined the impact of haptic feedback in handshake interactions, showing that sensing-based feedback improves the robot's reaction and user's emotional engagement [17]. The literature shows an interest in sensor-driven HRI in various SAR applications, including

sensor development and robotic instrumentation in different SAR areas. Moreover, a wider dataset is needed to provide varied data considering inter-subject forces and kinds of interactions. This leads to the opportunity to extend this work to other social scenarios (i.e., emotional support, elderly care, and education) and other robotic platforms, considering the adaptability of the FBG-based network developed.

As future work, the sensor network will be extended beyond the hand to include other parts of the robot's body, as well as including more users, expanding the variability of the dataset. Regions such as arms, trunk, and head are commonly involved in physical interactions and social gestures. Instrumenting these areas will expand the system's coverage and enable the detection and classification of a broader range of interaction types, which is essential for a more comprehensive and accurate analysis of physical human-robot interaction. Finally, on future work, more validation strategies (e.g., stratified cross-validation and leave-one-user-out) and ML algorithms variations (e.g., random forest and long short-term memory neural network) can be tested.

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